

Modeling of Mode I Delamination Growth in Composites by using Peridynamics Implemented in Abaqus

Uğur Yolum, Ahmet Taştan, Tezcan Kahraman, Mehmet A. Guler
Erkan Oterkus, Erdogan Madenci

TOBB University of Economics and Technology

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Content

- Short history
- Motivation
- Theory
- Implementation
- Problem Definitions and Results

Short history

- LEFM, Infinite stresses at crack tips (Griffith, 1921)
- Pioneering works of nonlocal theories (A.C. Eringen, 1972)
- Development of Peridynamic theory (Silling, 2000)
- Recent works (For example E. Madenci, 2013)

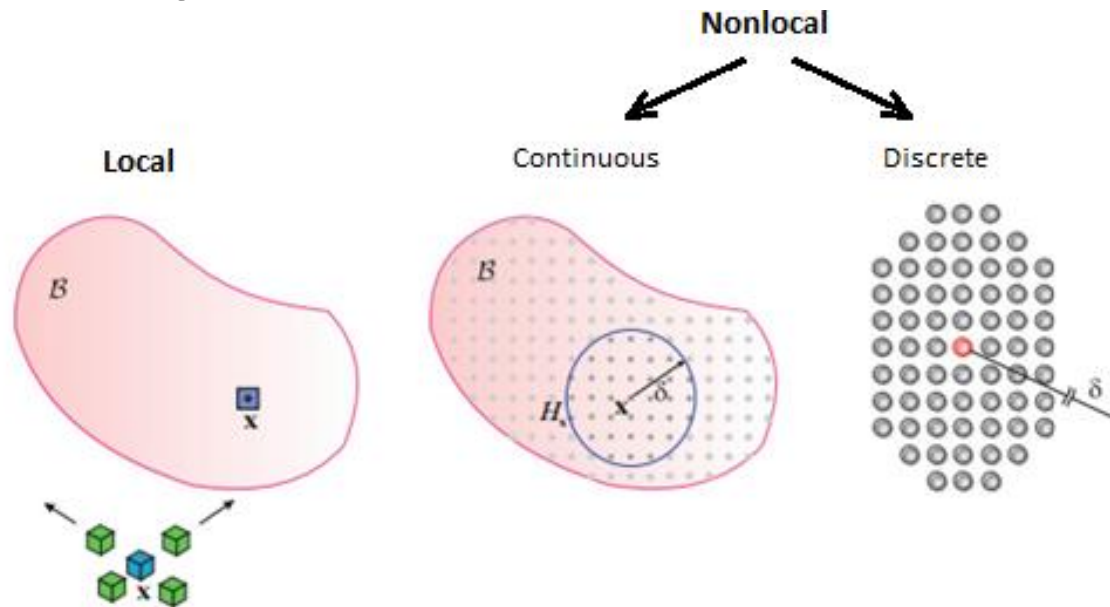
Motivation

Weaknesses in failure predictions in LEFM

- Need for pre-existing cracks
- Infinite stresses at crack tips
- Difficulty in accurate calculation of stress intensity factor and critical energy release rate

Motivation

Locality vs Nonlocality



Classical continuum mechanics

the state of a material point is influenced by the material points located in its immediate vicinity

Peridynamics

the state of a material point is influenced by material points located within a region of finite radius

Molecular dynamics

* Oterkus, E.: Peridynamic theory for modeling three-dimensional damage growth in metallic and composite structures, Ph. D. Thesis, The University of Arizona (2010)

Theory

Equation of Motion in Classical Continuum Mechanics:

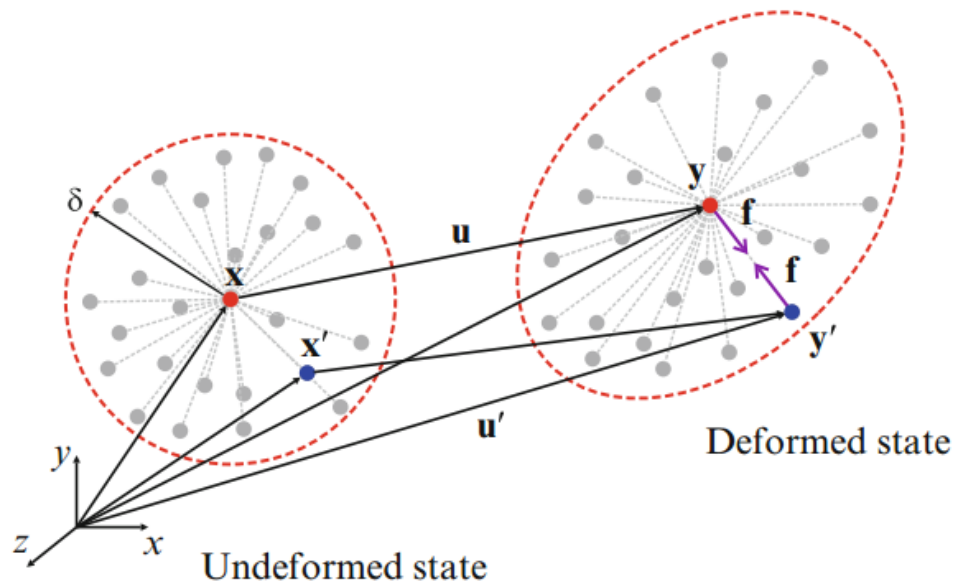
$$\rho(\mathbf{x})\ddot{\mathbf{u}} = \nabla \cdot \boldsymbol{\sigma} + \mathbf{b}(\mathbf{x}, t)$$

Peridynamic Equation of Motion:

$$\rho(\mathbf{x})\ddot{\mathbf{u}}(\mathbf{x}, t) = \int_H \mathbf{f}(\mathbf{u}' - \mathbf{u}, \mathbf{x}' - \mathbf{x}) dH + \mathbf{b}(\mathbf{x}, t)$$

Theory

Bond-based Peridynamics



$$\xi = \mathbf{x}' - \mathbf{x}$$

$$\boldsymbol{\eta} = \mathbf{u}(\mathbf{x}', t) - \mathbf{u}(\mathbf{x}, t)$$

$$\mathbf{y}' - \mathbf{y} = \boldsymbol{\eta} + \xi$$

Force density vector

$$\mathbf{f}(\boldsymbol{\eta}, \xi) = \frac{\xi + \boldsymbol{\eta}}{|\xi + \boldsymbol{\eta}|} c_s \quad (\text{Pairwise interaction})$$

Stretch between material points :

$$s = \frac{|\xi + \boldsymbol{\eta}| - |\xi|}{|\xi|}$$

Theory

Material constant (c) derivation for a two dimensional plate

- Isotropic expansion loading, ($s = \zeta$)

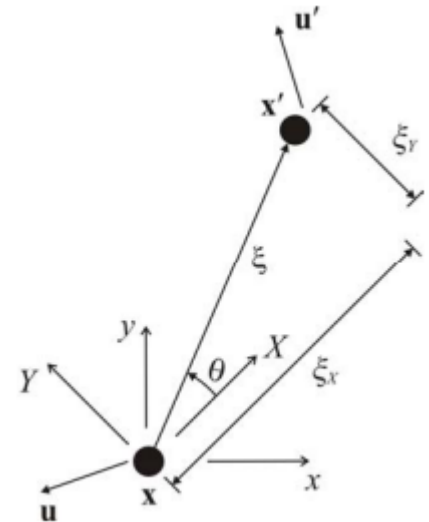
Strain energy density in Peridynamics:

$$U^{PD} = \int_H w dH \quad w = \frac{1}{2} c s^2 \xi$$

Strain energy density in Classical continuum mechanics:

$$U^{CCM} = \frac{E}{1-\nu} \zeta^2$$

Material constant (c) :
$$c = \frac{9E}{\pi t \delta^3}$$



Theory

Material constants of a lamina

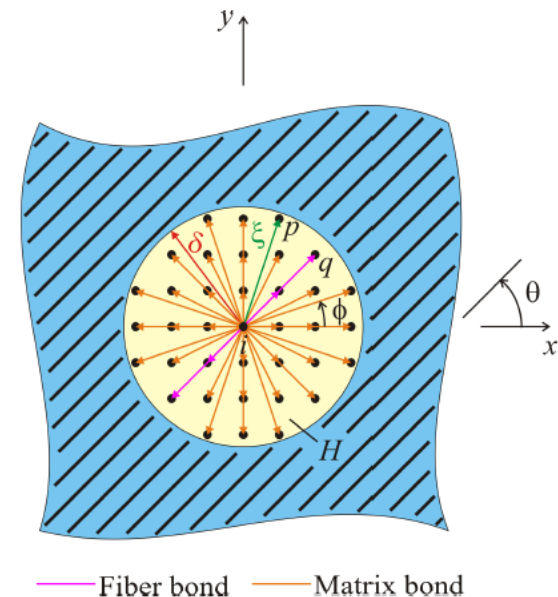
Micropotential for a laminate is : $w = \frac{1}{2} c(\phi) s(\phi)^2 \xi$

Strain energy density in Classical continuum mechanics

Material constants (c_f and c_m) : $U^{CCM} = \frac{1}{2} (\bar{\mathbf{Q}} \boldsymbol{\varepsilon})^T \boldsymbol{\varepsilon}$

$$c_f = \frac{Q_{11} - Q_{22}}{\frac{1}{2} \sum_{q=1}^Q \xi_{qi} V_q}$$

$$c_m = \frac{8Q_{22}}{\pi t \delta^3}$$



Theory

Material constants of a laminate

Strain energy density for interlayer bonds

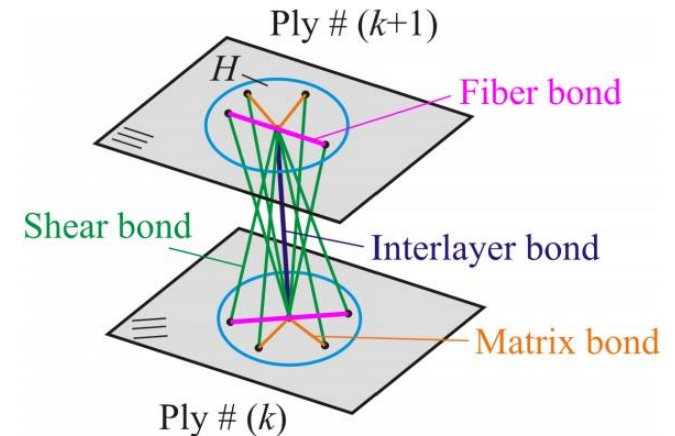
$$U^{PD} = \frac{c_{in} \zeta^2 t \bar{V}}{2}$$

Strain energy density in Classical continuum mechanics

Material constants (c_f and c_m) : $U^{CCM} = \frac{1}{2} E_m \zeta^2$

$$c_{in} = \frac{E_m}{t \bar{V}}$$

E_m : Elastic modulus of matrix material

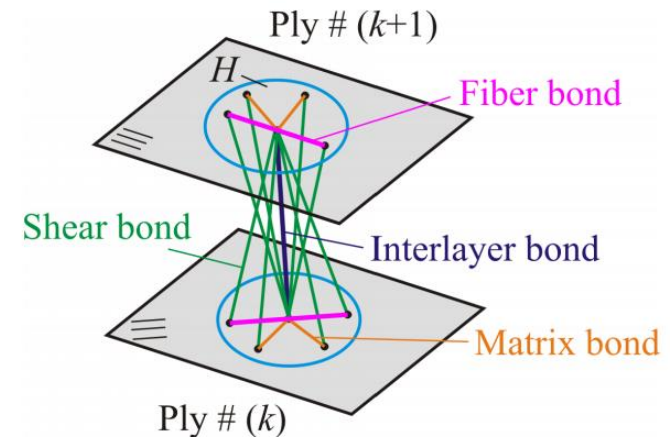


Theory

Material constants of a laminate

Strain energy density for shear bonds

$$U^{PD} = \left[\frac{\pi C_s t}{4} (\delta^2 + t^2 \ln \frac{t^2}{t^2 + \delta^2}) \right] \zeta^2$$



Strain energy density in Classical continuum mechanics

$$U^{CCM} = \frac{1}{2} G_m \zeta^2$$

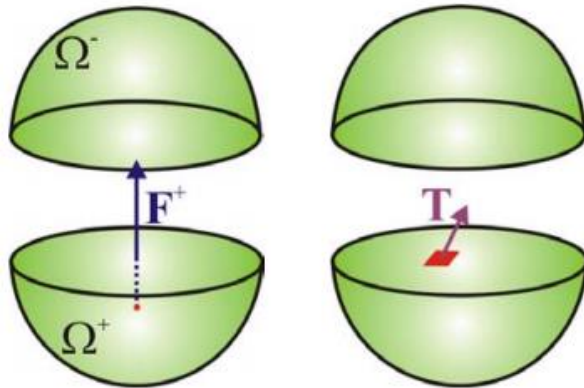
Material constants (c_s) :

$$c_s = \frac{2G_m}{\pi t} \frac{1}{\left[\delta^2 + t^2 \ln \left(\frac{t^2}{\delta^2 + t^2} \right) \right]}$$

G_m : Shear modulus of matrix material

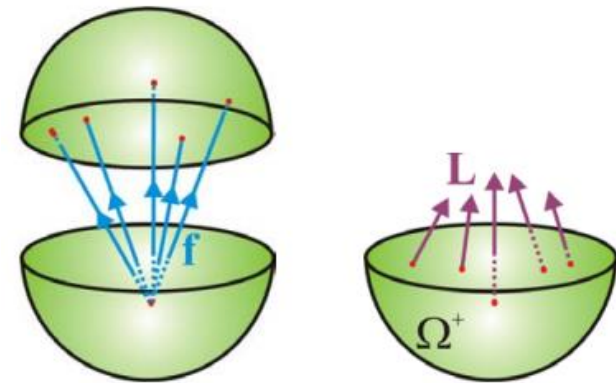
Theory

Application of Boundary Conditions



a) Classical Continuum Theory

- A net force, $\mathbf{F}^+$
- Surface tractions, $\mathbf{T}$
- Integrating tractions over the cross sectional area



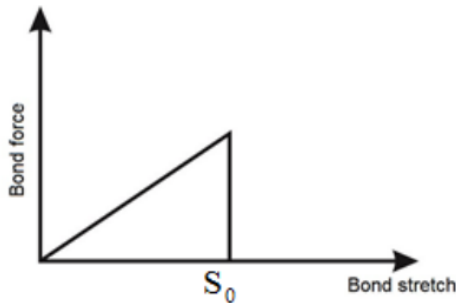
b) Peridynamic Theory

- Interaction of a point, $\mathbf{f}$
- Force densities, $\mathbf{L}$
- Volume integration of force densities

Theory

Material Failure in Peridynamics

- Adding failure parameter: $\mathbf{f}(\boldsymbol{\eta}, \xi) = \mu(\mathbf{x}' - \mathbf{x}, t) \frac{\xi + \boldsymbol{\eta}}{|\xi + \boldsymbol{\eta}|} c s$

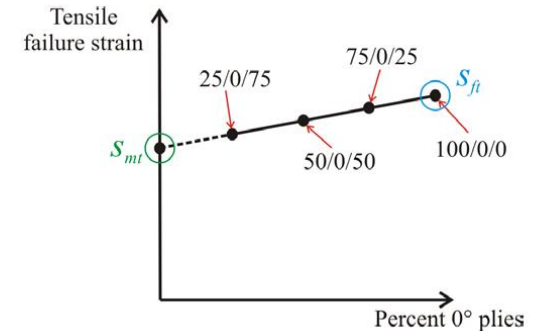


Brittle materials

$$s_m = \sqrt{\frac{5G_{IC}}{9\kappa_m\delta}}$$

$$s_{in} = \sqrt{\frac{2G_{IC}}{tE_m}}$$

$$\psi_c = \sqrt{\frac{G_{IIC}}{tG_m}}$$



s_0 : The critical stretch value

- Interaction between two points: $\mu(\mathbf{x}' - \mathbf{x}, t) = \begin{cases} 1 & , s(\mathbf{x}' - \mathbf{x}, t) < s_0 \\ 0 & , s(\mathbf{x}' - \mathbf{x}, t) > s_0 \end{cases}$
- The critical stretch value is related to critical energy release rate (G)

Implementation

- Discretized form of equation of motion:

$$\rho \ddot{\mathbf{u}}_i^n = \sum_p f(\mathbf{u}_p^n - \mathbf{u}_i^n, \mathbf{x}_p - \mathbf{x}_i) V_p + \mathbf{b}_i^n$$

- Multiplying equation of motion by V_i :

$$V_i \rho \ddot{\mathbf{u}}_i^n = \sum_p f(\mathbf{u}_p^n - \mathbf{u}_i^n, \mathbf{x}_p - \mathbf{x}_i) V_p V_i + \mathbf{b}_i^n V_i$$

- Stiffness properties (Macek and Silling 2007):

$$\begin{aligned} A_t &= (V_p V_i)^{1/3} & E_t &= c (V_p V_i)^{2/3} \\ A_t &= \Delta x^2 & E_t &= c \Delta x^4 \quad (\text{If thickness} = \Delta x \text{ for a 2D Plate}) \end{aligned}$$

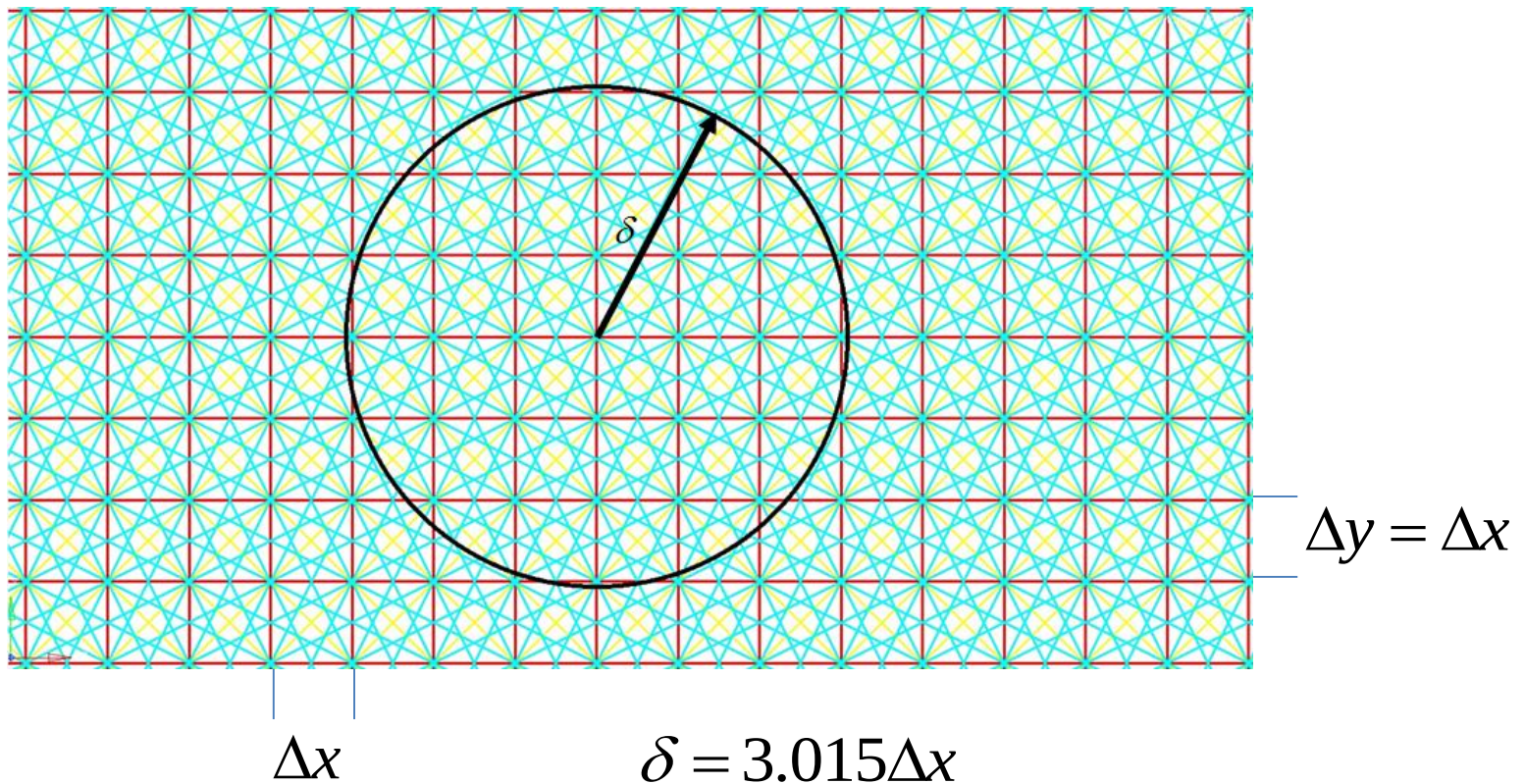
$$c = \frac{9E}{\pi t \delta^3}$$

- Fracture strain of the truss:

$$\varepsilon_f = s_0$$

Implementation

Representation of Peridynamic bonds, truss elements



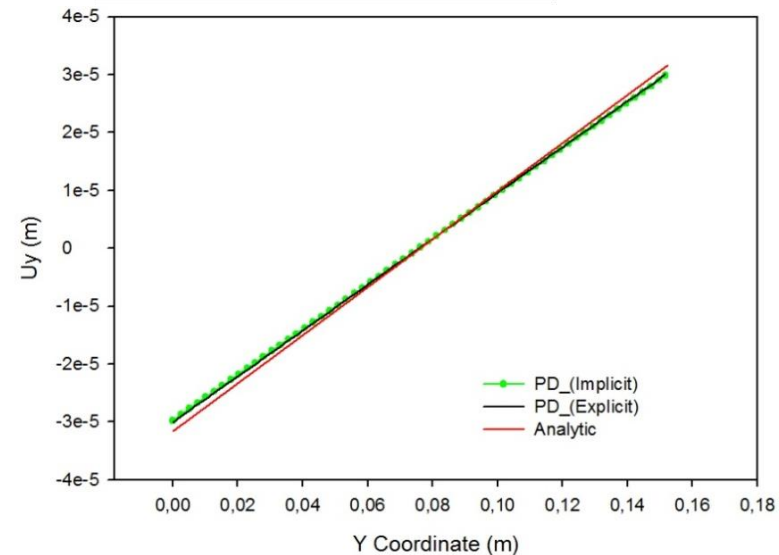
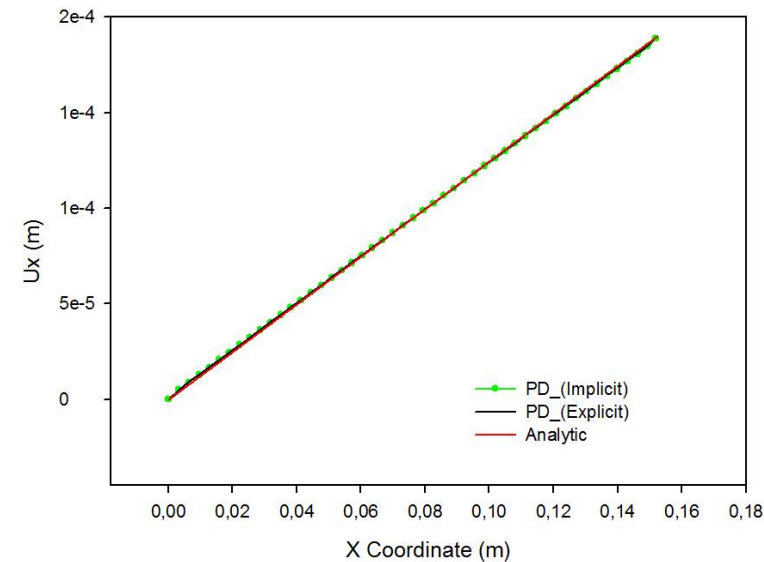
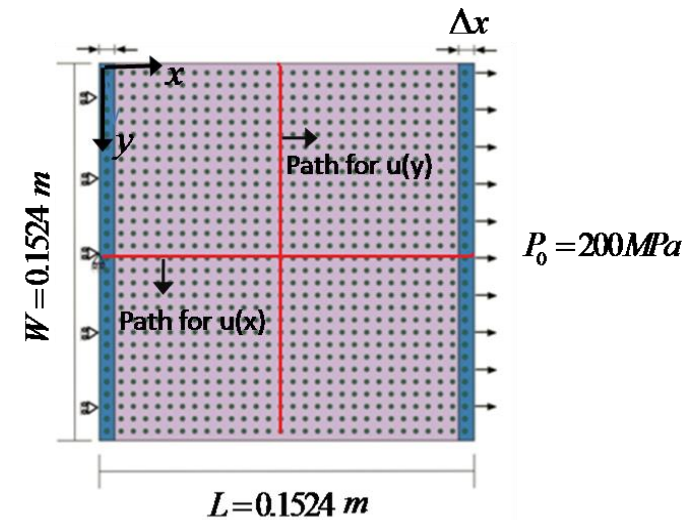
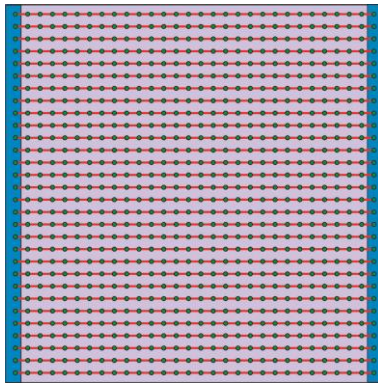
Material Properties

- Bond properties are derived using material data below:

E11 [GPa]	152
E22 [GPa]	8.52
ν	1/3
GIC [N/m]	352.3

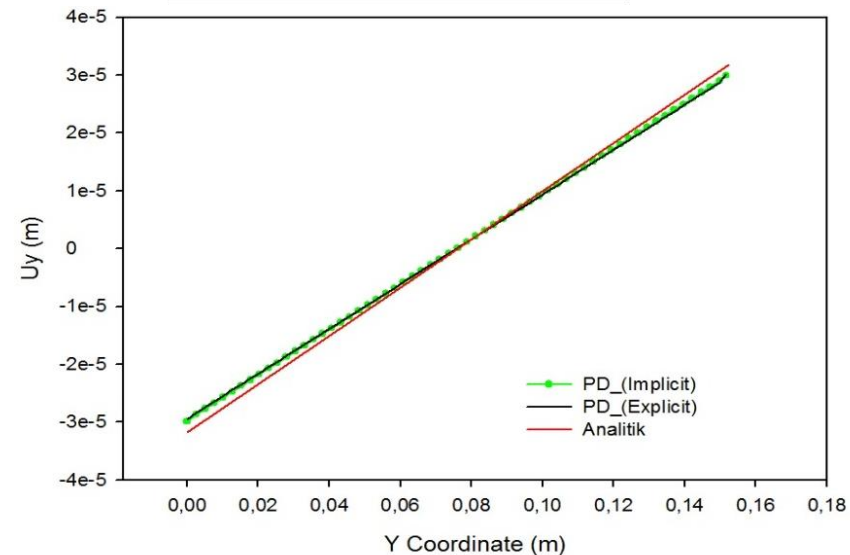
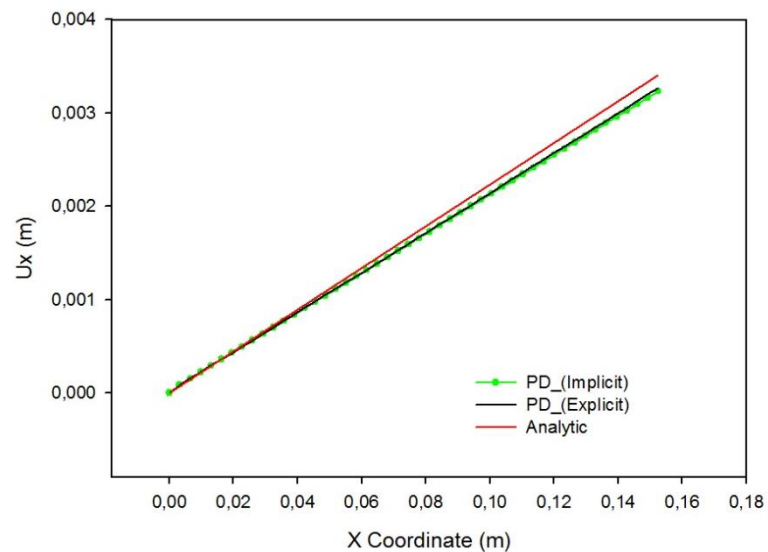
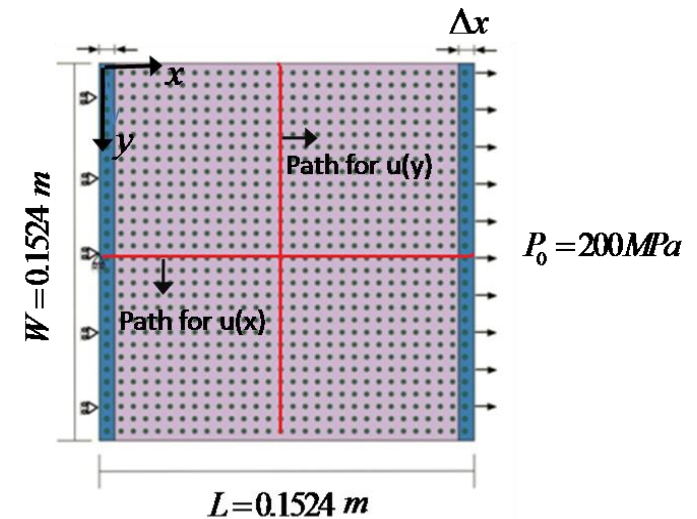
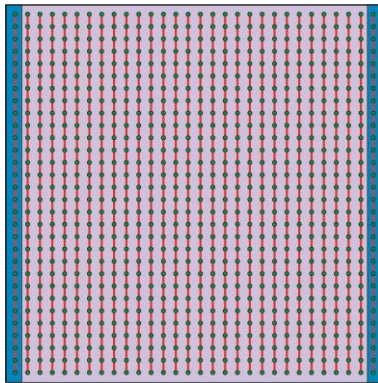
Lamina Under Tension

0^0 Ply

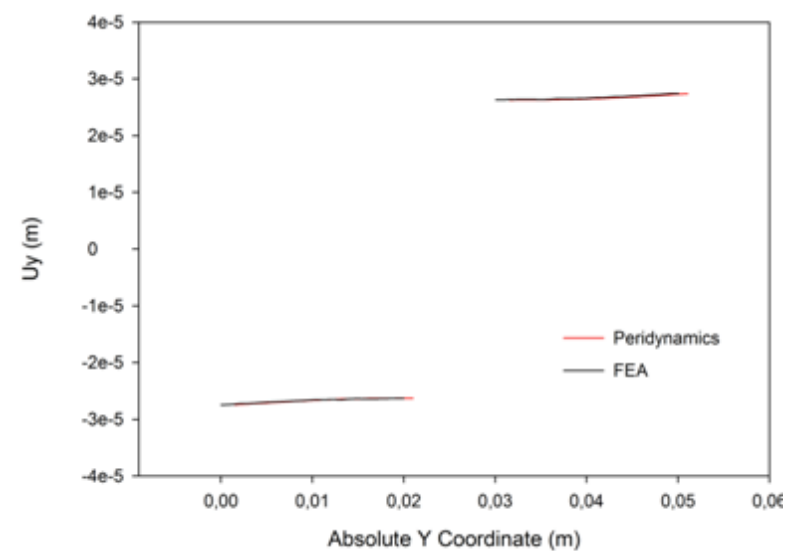
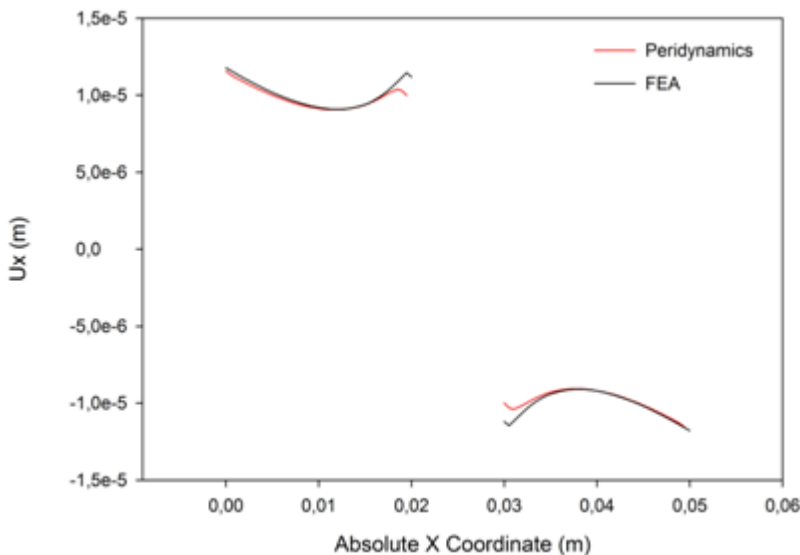
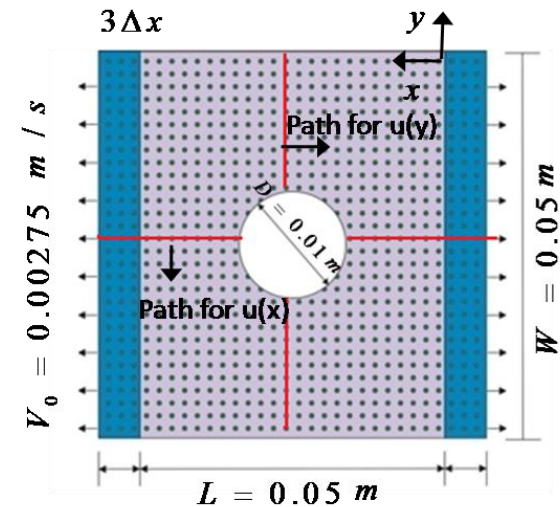
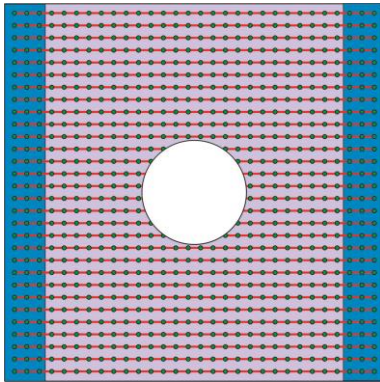


Lamina Under Tension

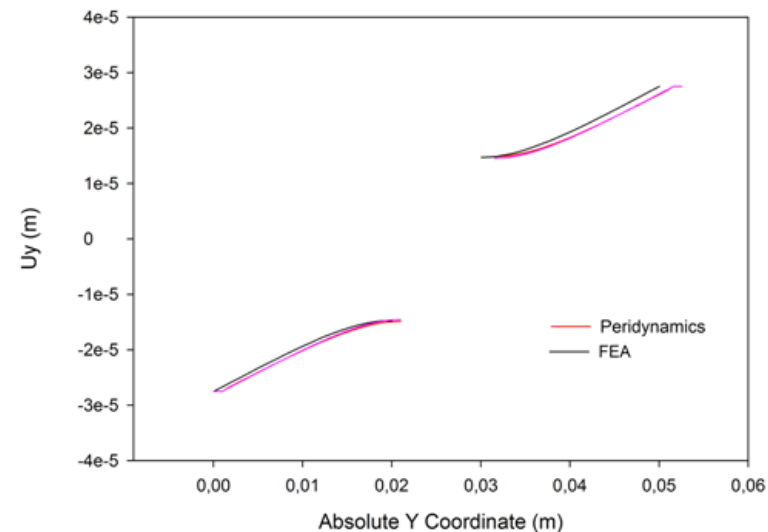
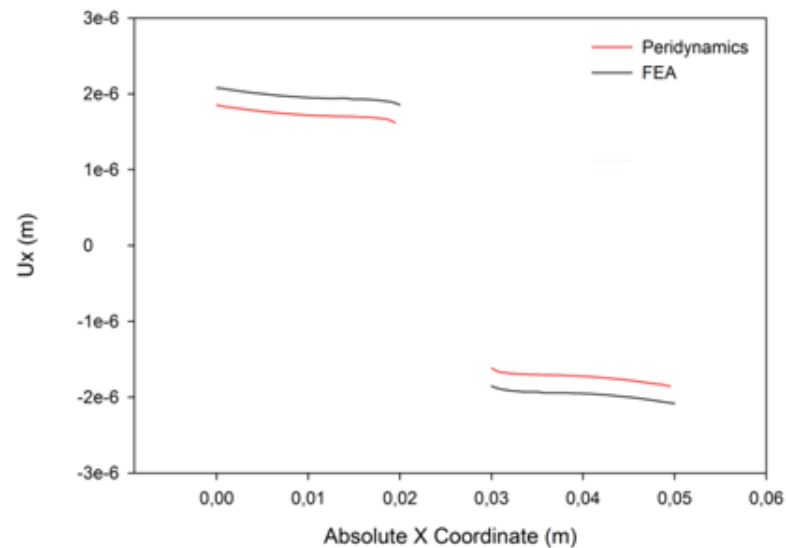
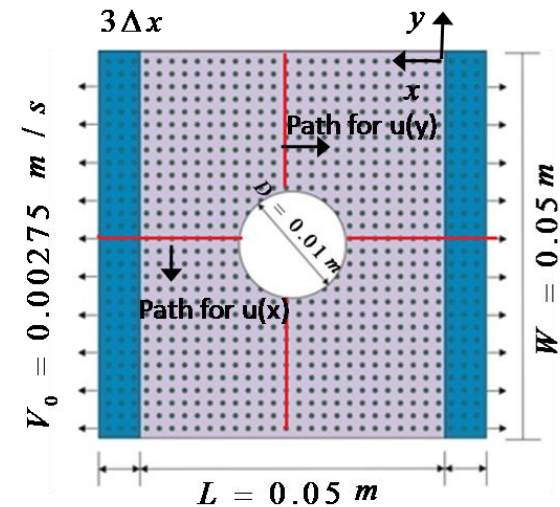
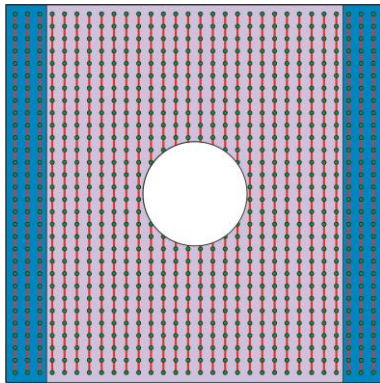
90° Ply



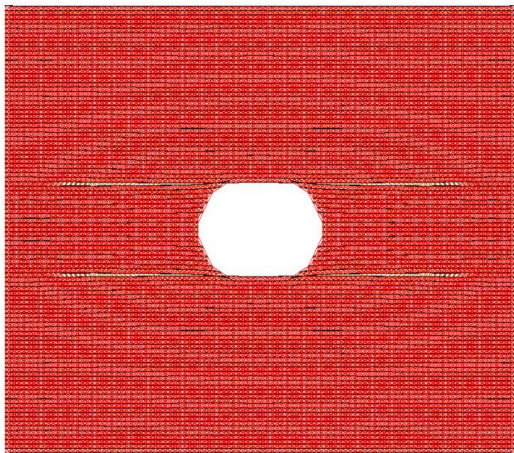
0° Lamina with a Circular Cutout Under Tension



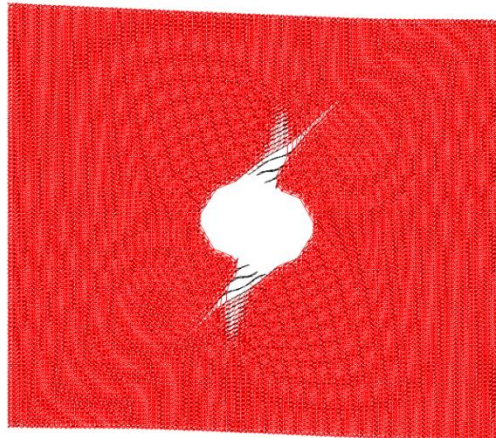
90° Lamina with a Circular Cutout Under Tension



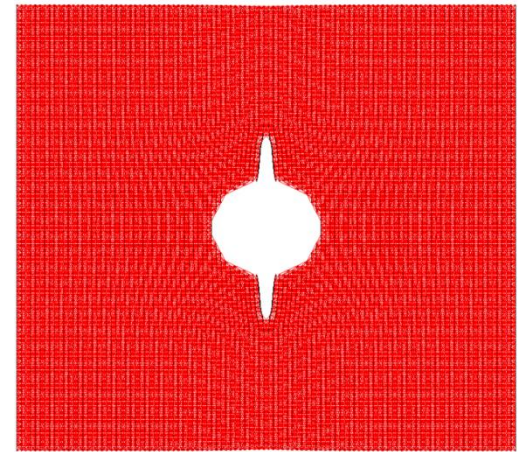
Lamina Under Tension (Damage model)



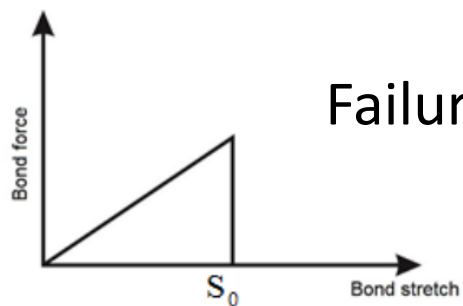
a) 0°



b) 45°



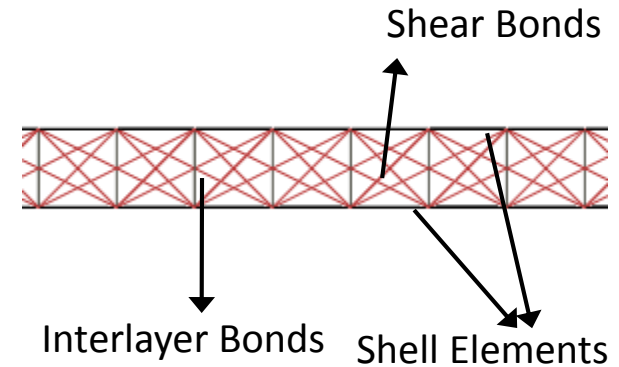
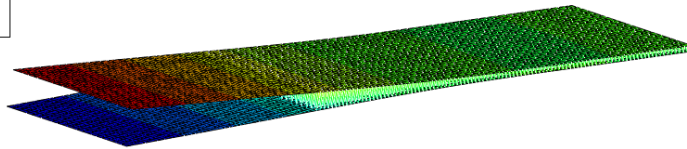
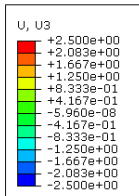
c) 90°



Failure propagation in 0° , 45° and 90° plies

DCB Experiment

- Plies are simulated with orthotropic shell elements
- Only interface properties are modeled with peridynamic approach
- The results are not validated yet



Specimen Width= 25 mm
Ply Thickness=2.0 mm

L=150 mm

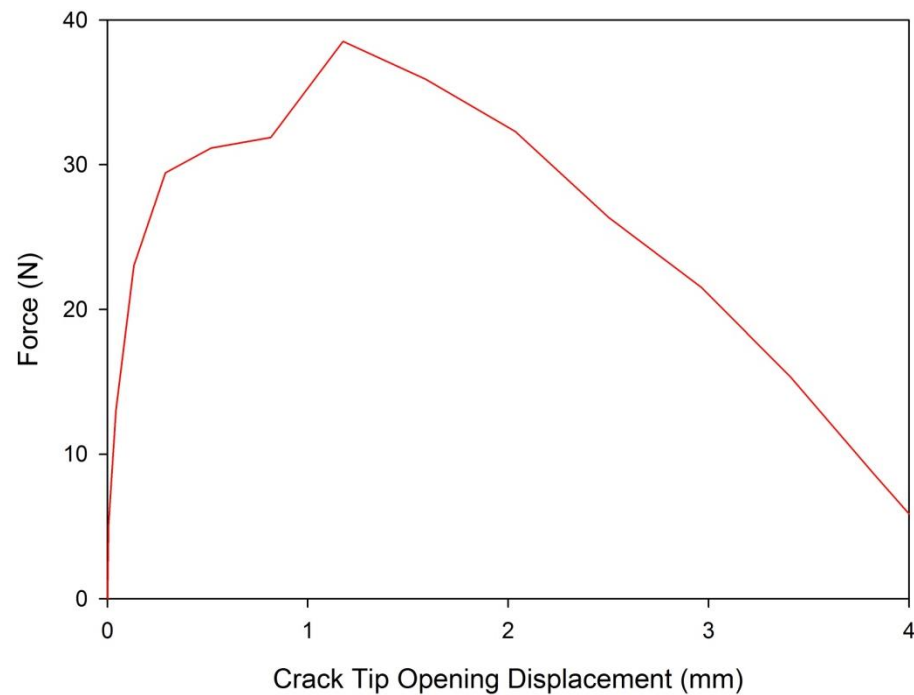
a=50 mm

u

u

Results

- DCB results are not yet validated.



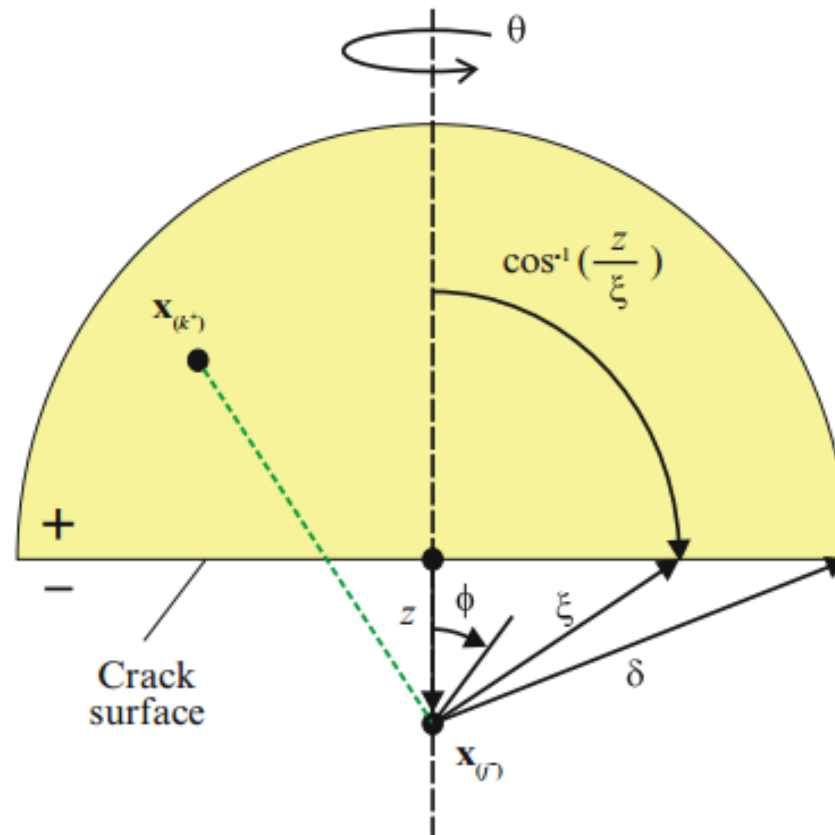
Acknowledgment

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THANKS FOR LISTENING!

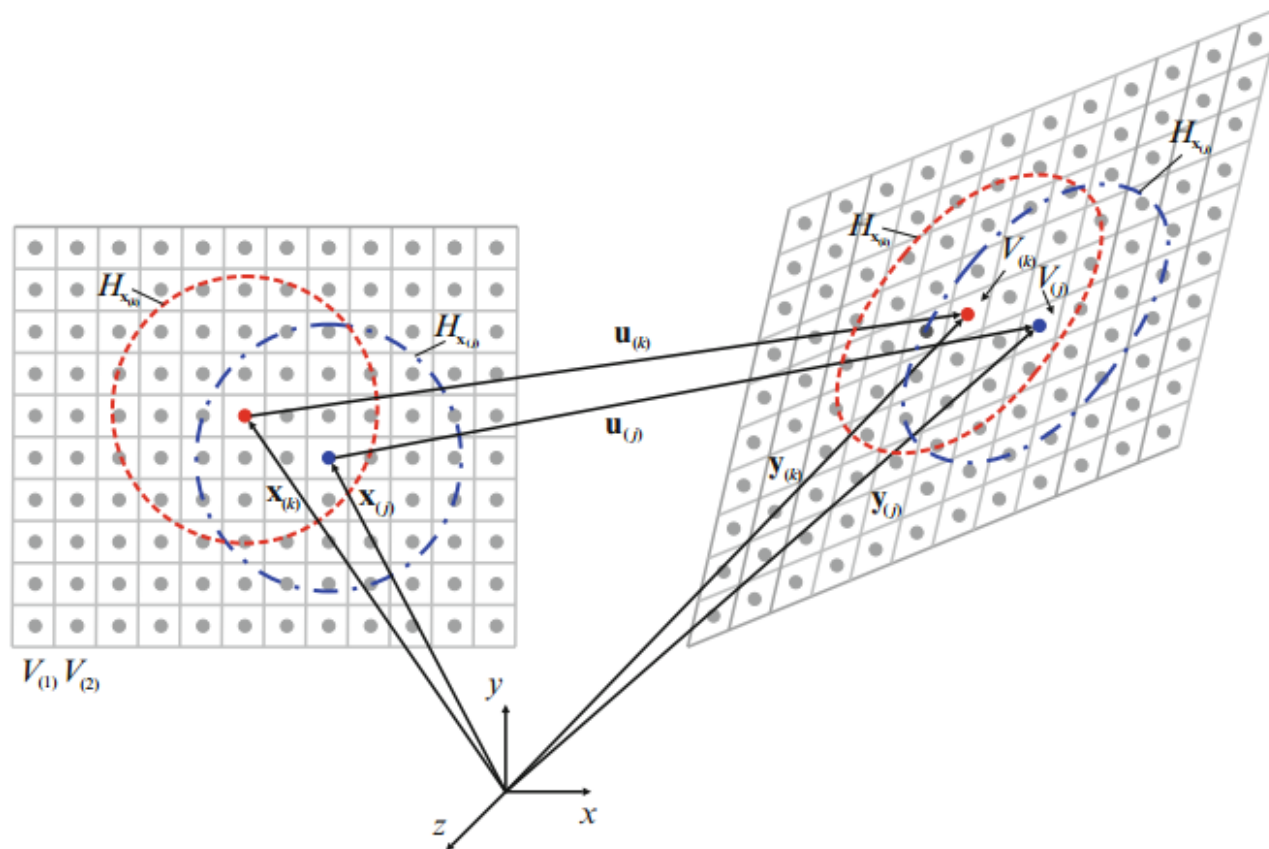
Theory

Integration domain of the micropotentials crossing a fracture surface



Theory

Kinematics of Peridynamic Material Points





Theory

Fig. 4.7 Surface effects in the domain of interest

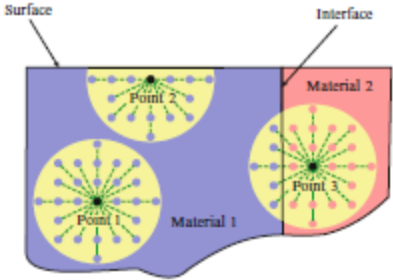


Fig. 4.10 Construction of an ellipsoid for surface correction factors

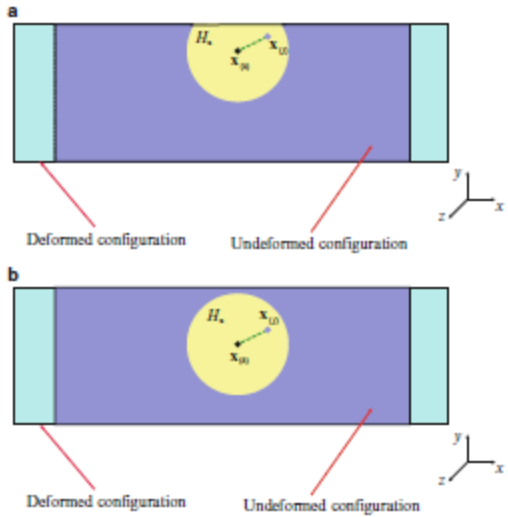
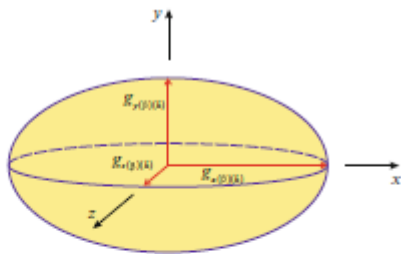
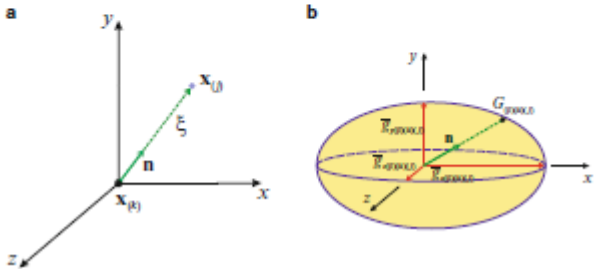
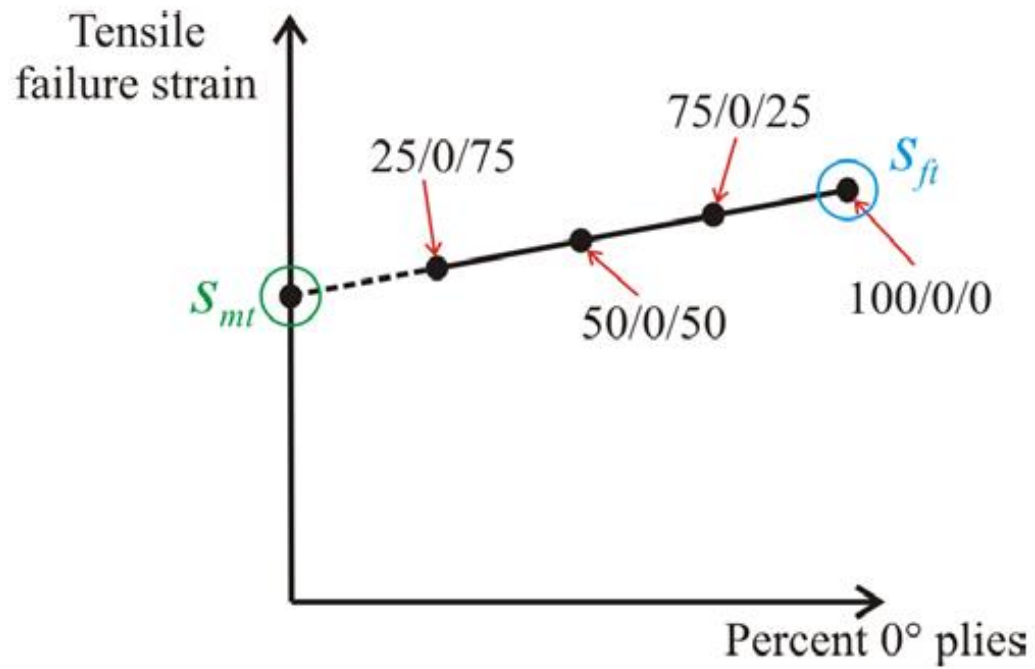


Fig. 4.8 Material point x with (a) a truncated horizon and (b) far away from external surfaces of a material domain subjected to uniaxial stretch loading







$$c = \frac{18k}{\pi\delta^4}. \quad (23)$$

The critical stretch for bond failure, s_0 , may also be related to a measurable quantity by considering a planar fracture surface in the interior of a large homogeneous body. To completely separate the two halves of the body across the fracture surface requires breaking all the bonds that initially connected points in the opposite halves. Let $w_0(\xi)$ denote the work required to break a single bond,

$$w_0(\xi) = \int_0^{s_0} g(s)(\xi ds), \quad \xi = |\xi|, \quad (24)$$

where the relation $d\eta = \xi ds$ has been used. In the present case of the PMB material, (24) leads to $w_0(\xi) = cs_0^2\xi/2$. The work G_0 required to break all the bonds per unit fracture area is then found from

$$G_0 = \int_0^\delta \int_0^{2\pi} \int_z^\delta \int_0^{\cos^{-1}z/\xi} (cs_0^2\xi/2)\xi^2 \sin\phi d\phi d\xi d\theta dz. \quad (25)$$

(Refer to Fig. 3 for an explanation of this computation.) After evaluation of these integrals, the energy per unit fracture area for complete separation of the two halves of the body is found to be

$$G_0 = \frac{\pi cs_0^2\delta^5}{10}. \quad (26)$$